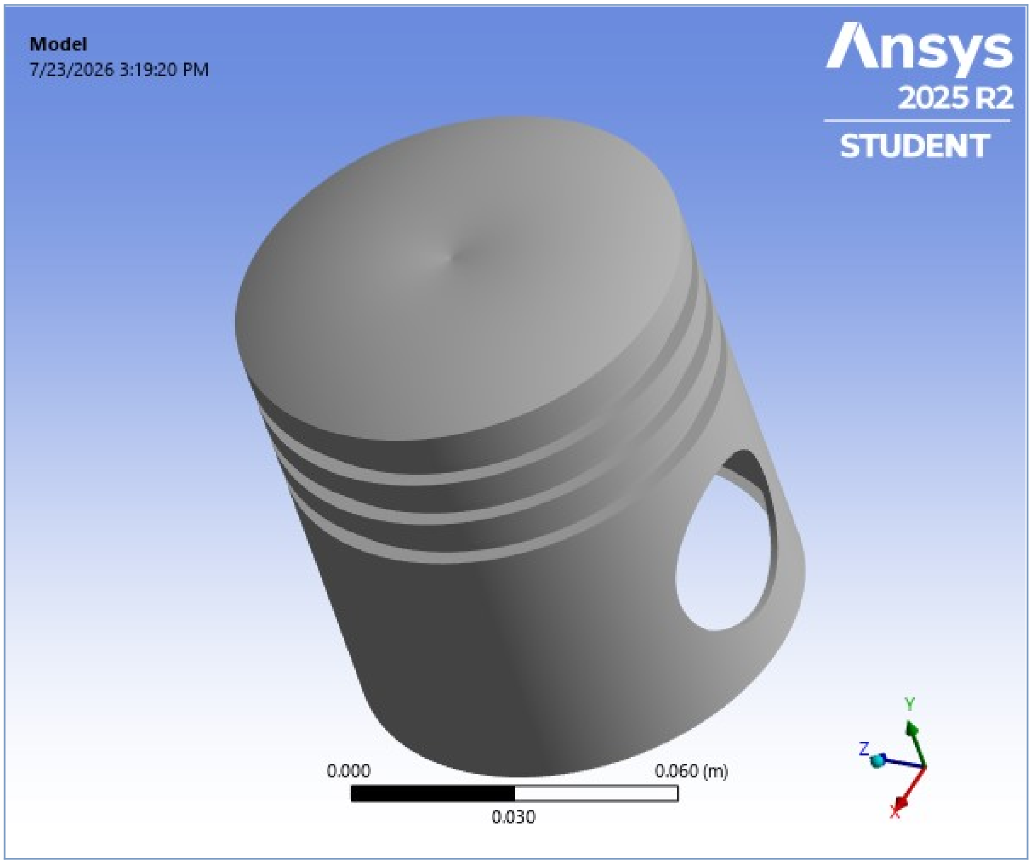
Project

First Saved	Thursday, July 23, 2026
Last Saved	Thursday, July 23, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (A4, B4)**
  - Geometry Imports
    - Geometry Import (A3, B3)
  - Geometry
    - SYS-2\Solid
  - Materials
  - Coordinate Systems
  - Mesh
  - **Transient Thermal (A5)**
    - Initial Temperature
    - Analysis Settings
    - Loads
    - Solution (A6)
      - Solution Information
        - Result Charts
      - Results
  - **Static Structural (B5)**
    - Analysis Settings
    - Loads
    - Imported Load (A6)
      - Imported Body Temperature
    - Solution (B6)
      - Solution Information
      - Results
- **Material Data**
  - Al2618

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, B4)

TABLE 2
Model (A4, B4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4, B4) > Geometry Imports > Geometry Import (A3, B3)

Object Name	*Geometry Import (A3, B3)*
State	Solved
Definition	
Source	C:\Users\User\OneDrive\Desktop\Piston\Ansys\Ansys\Completefile\Complete_files\dp0\SYS-2\DM\SYS-2.scdocx
Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes

Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4, B4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\User\OneDrive\Desktop\Piston\Ansys\Ansys\Completefile\Complete_files\dp0\SYS-2\DM\SYS-2.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	8.5e-002 m
Length Y	9.238e-002 m
Length Z	8.5e-002 m
Properties	
Volume	4.6047e-005 m³
Mass	0.12755 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	16310
Elements	8039
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes

Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4, B4) > Geometry > Parts

Object Name	*SYS-2\Solid*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Al2618
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	8.5e-002 m
Length Y	9.238e-002 m
Length Z	8.5e-002 m
Properties	
Volume	4.6047e-005 m³
Mass	0.12755 kg
Centroid X	3.7474e-006 m
Centroid Y	5.5486e-002 m
Centroid Z	8.8314e-009 m
Moment of Inertia Ip1	1.8909e-004 kg·m²
Moment of Inertia Ip2	1.9508e-004 kg·m²
Moment of Inertia Ip3	2.0127e-004 kg·m²
Statistics	
Nodes	16310
Elements	8039
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:175.143.175	

TABLE 6
Model (A4, B4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4, B4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Mesh

TABLE 8
Model (A4, B4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	0.1516 m
Average Surface Area	1.7877e-003 m²
Minimum Edge Length	2.4869e-002 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5

Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	16310
Elements	8039
Show Detailed Statistics	No

Transient Thermal (A5)

TABLE 9
Model (A4, B4) > Analysis

Object Name	*Transient Thermal (A5)*
State	Solved
Definition	
Physics Type	Thermal
Analysis Type	Transient
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 10
Model (A4, B4) > Transient Thermal (A5) > Initial Condition

Object Name	*Initial Temperature*
State	Fully Defined
Definition	
Initial Temperature	Uniform Temperature
Initial Temperature Value	22. °C

TABLE 11
Model (A4, B4) > Transient Thermal (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Initial Time Step	1.e-002 s
Minimum Time Step	1.e-003 s
Maximum Time Step	0.1 s
Time Integration	On
Solver Controls	
Solver Type	Program Controlled
Radiosity Controls	
Radiosity Solver	Program Controlled
Flux Convergence	1.e-004
Maximum Iteration	1000.

Solver Tolerance	0.1 W/m²
Over Relaxation	0.1
Hemicube Resolution	10.
Nonlinear Controls	
Heat Convergence	Program Controlled
Temperature Convergence	Program Controlled
Line Search	Program Controlled
Nonlinear Formulation	Program Controlled
Advanced	
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Calculate Thermal Flux	Yes
Contact Data	Yes
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\User\OneDrive\Desktop\Piston\Ansys\Ansys\d\d_files\dp0\SYS-2\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	mks

TABLE 12
Model (A4, B4) > Transient Thermal (A5) > Loads

Object Name	*Temperature*	*Convection*	*Radiation*
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	3 Faces	13 Faces	16 Faces
Definition			
Type	Temperature	Convection	Radiation
Magnitude	450. °C (step applied)		
Suppressed	No		
Film Coefficient		Tabular Data	
Ambient Temperature		Tabular Data	150. °C (step applied)
Convection Matrix		Program Controlled	
Element APDL Name			
Edit Data For		Film Coefficient	
Correlation			To Ambient
Emissivity			0.5 (step applied)
Tabular Data			
Independent Variable		Time	

FIGURE 1
Model (A4, B4) > Transient Thermal (A5) > Temperature

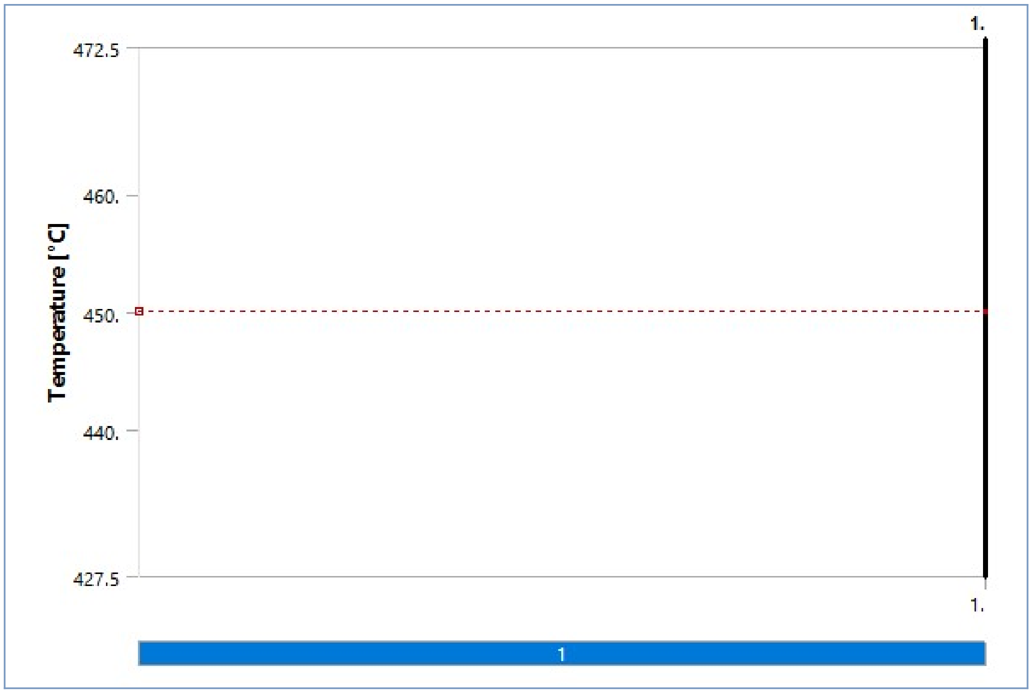


FIGURE 2
Model (A4, B4) > Transient Thermal (A5) > Convection

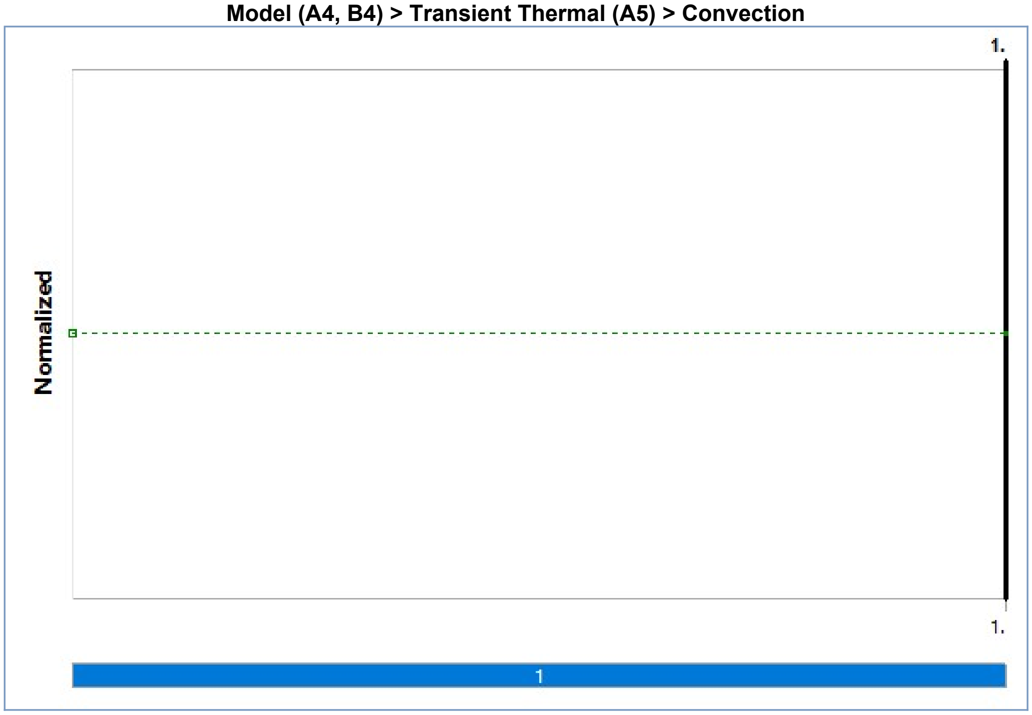
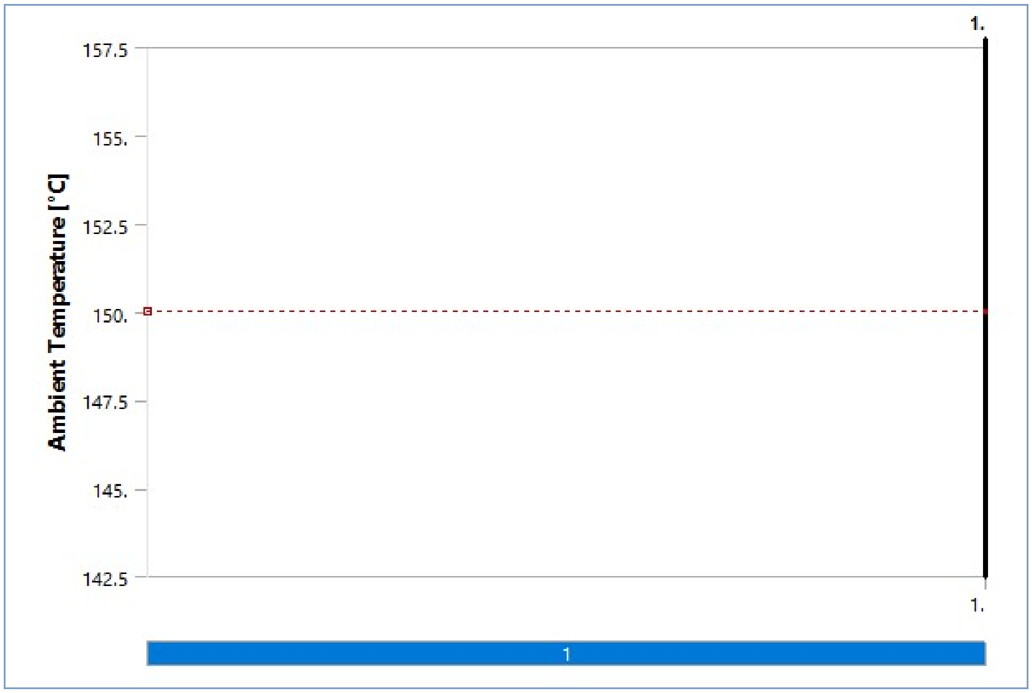


TABLE 13
Model (A4, B4) > Transient Thermal (A5) > Convection

Steps	Time [s]	Convection Coefficient [W/m²·°C]	Temperature [°C]
1	0.	= 1500.	= 90.
	1.	1500.	90.

FIGURE 3
Model (A4, B4) > Transient Thermal (A5) > Radiation



Solution (A6)

TABLE 14
Model (A4, B4) > Transient Thermal (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	33. s
MAPDL Memory Used	199. MB
MAPDL Result File Size	30.312 MB
Post Processing	
Beam Section Results	No

TABLE 15
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 16
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Result Charts

Object Name	*Temperature - Global Maximum*	*Temperature - Global Minimum*
State	Solved	

Scope		
Scoping Method	Global Maximum	Global Minimum
Definition		
Type	Temperature	
Suppressed	No	
Results		
Minimum	452.53 °C	90.382 °C
Maximum	452.53 °C	90.382 °C

FIGURE 4
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Maximum

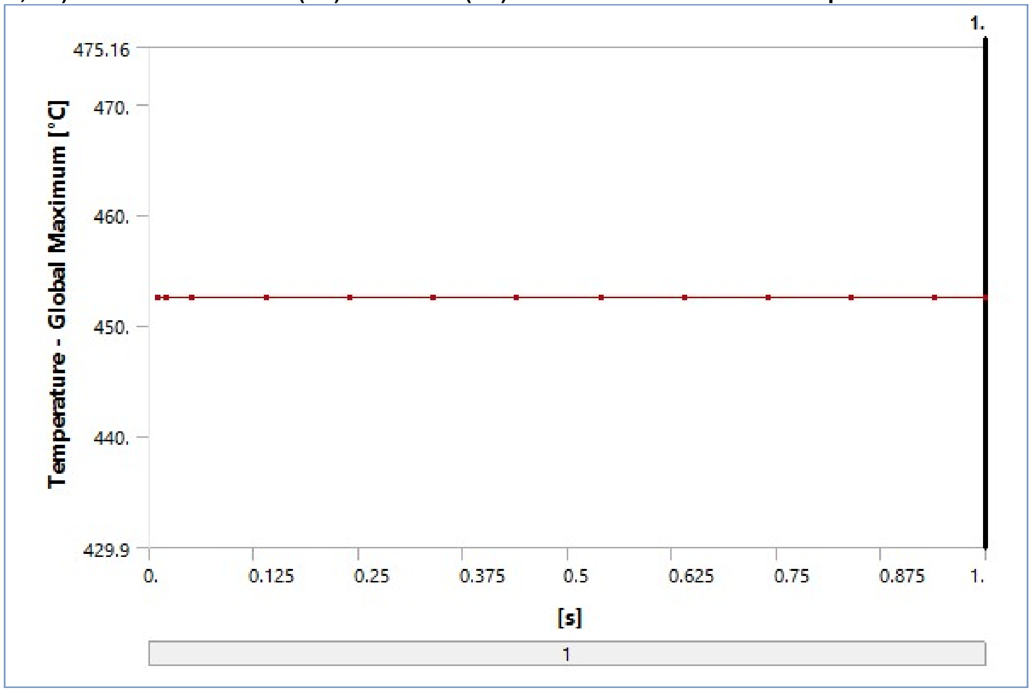


FIGURE 5
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Minimum

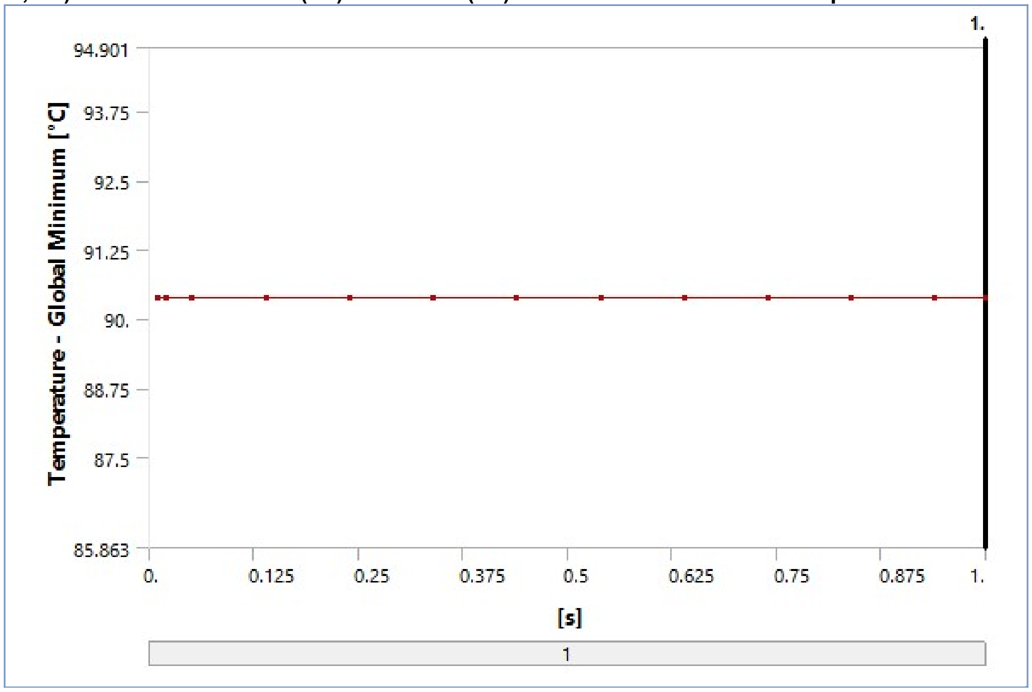


TABLE 17
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Results

Object Name	*Temperature*	*Total Heat Flux*
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Temperature	Total Heat Flux
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Results		
Minimum	90.382 °C	1.4419e-004 W/m²
Maximum	452.53 °C	4.4084e+006 W/m²
Average	185.6 °C	6.1577e+005 W/m²
Minimum Occurs On	SYS-2\Solid	
Maximum Occurs On	SYS-2\Solid	
Minimum Value Over Time		
Minimum	90.382 °C	1.4419e-004 W/m²
Maximum	90.382 °C	1.4419e-004 W/m²
Maximum Value Over Time		
Minimum	452.53 °C	4.4084e+006 W/m²
Maximum	452.53 °C	4.4084e+006 W/m²
Information		
Time	1. s	
Load Step	1	
Substep	13	
Iteration Number	14	
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 6
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Temperature

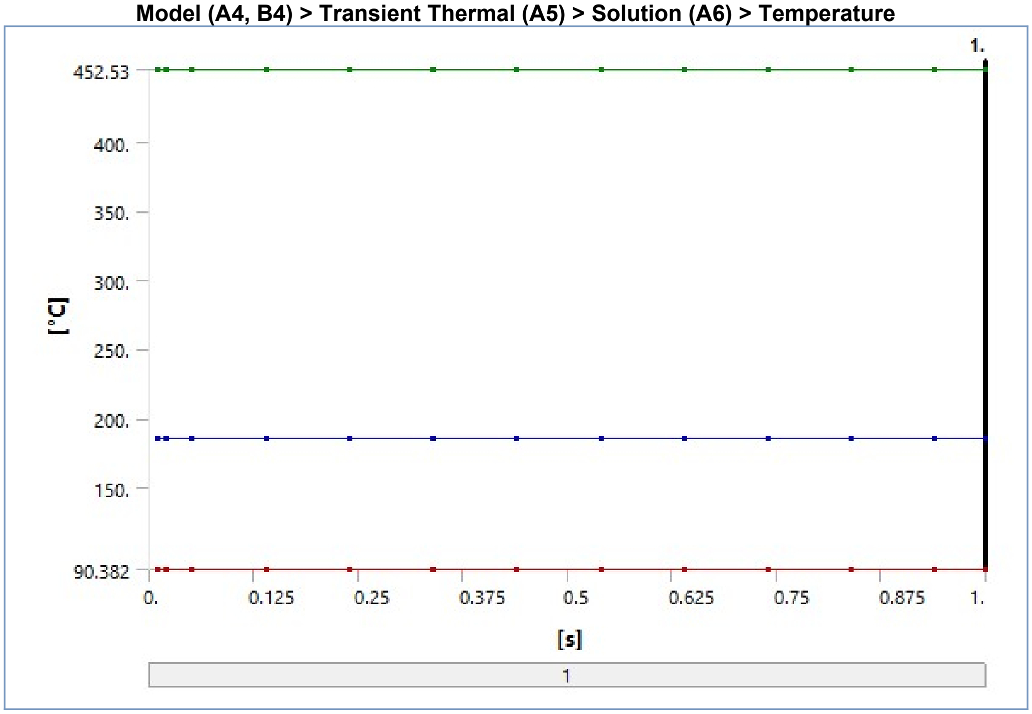


TABLE 18
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Temperature

Time [s]	Minimum [°C]	Maximum [°C]	Average [°C]
1.e-002	90.382	452.53	185.6
2.e-002			
5.e-002			
0.14			
0.24			
0.34			
0.44			
0.54			
0.64			
0.74			
0.84			
0.94			
1.			

FIGURE 7
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Temperature > Figure

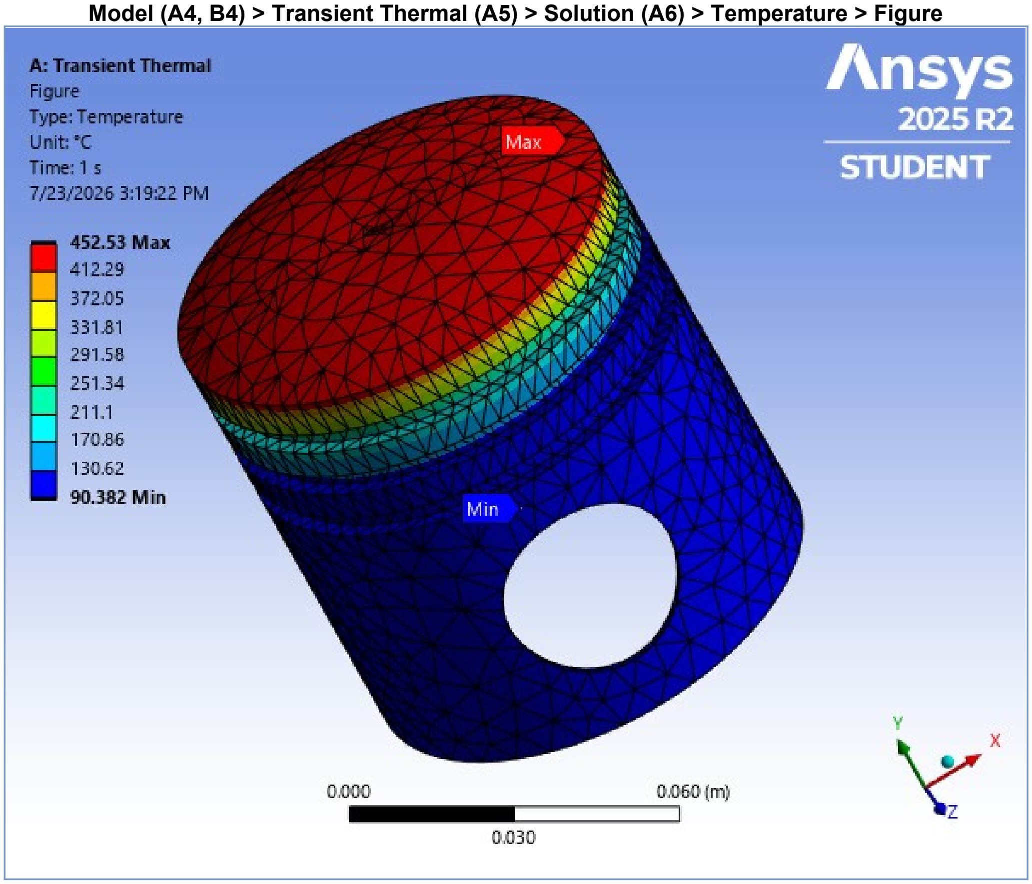


FIGURE 8
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux

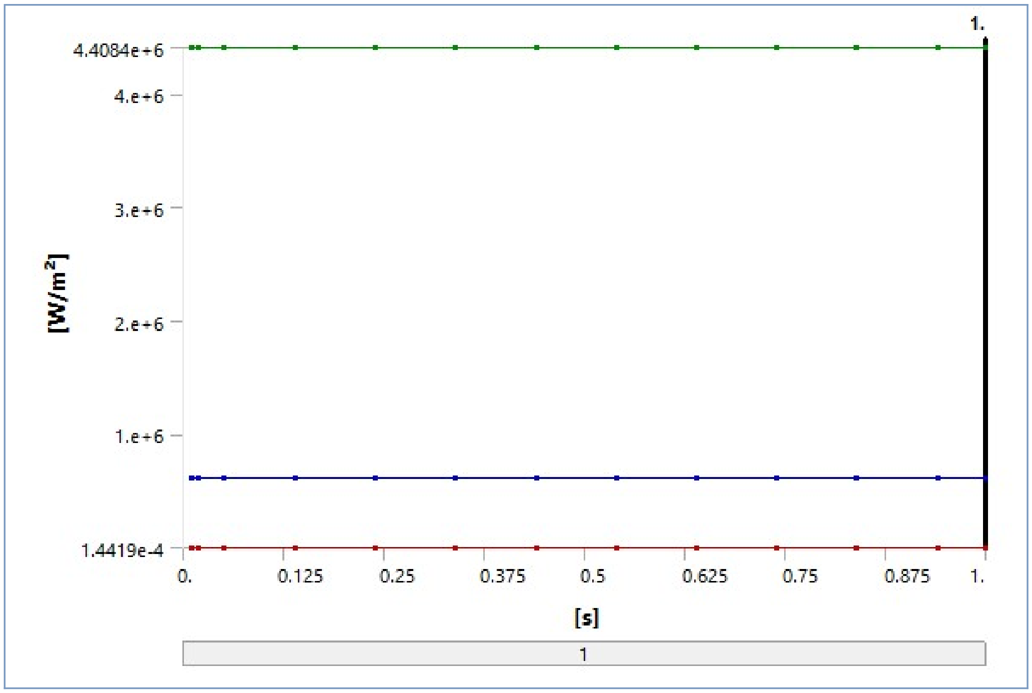
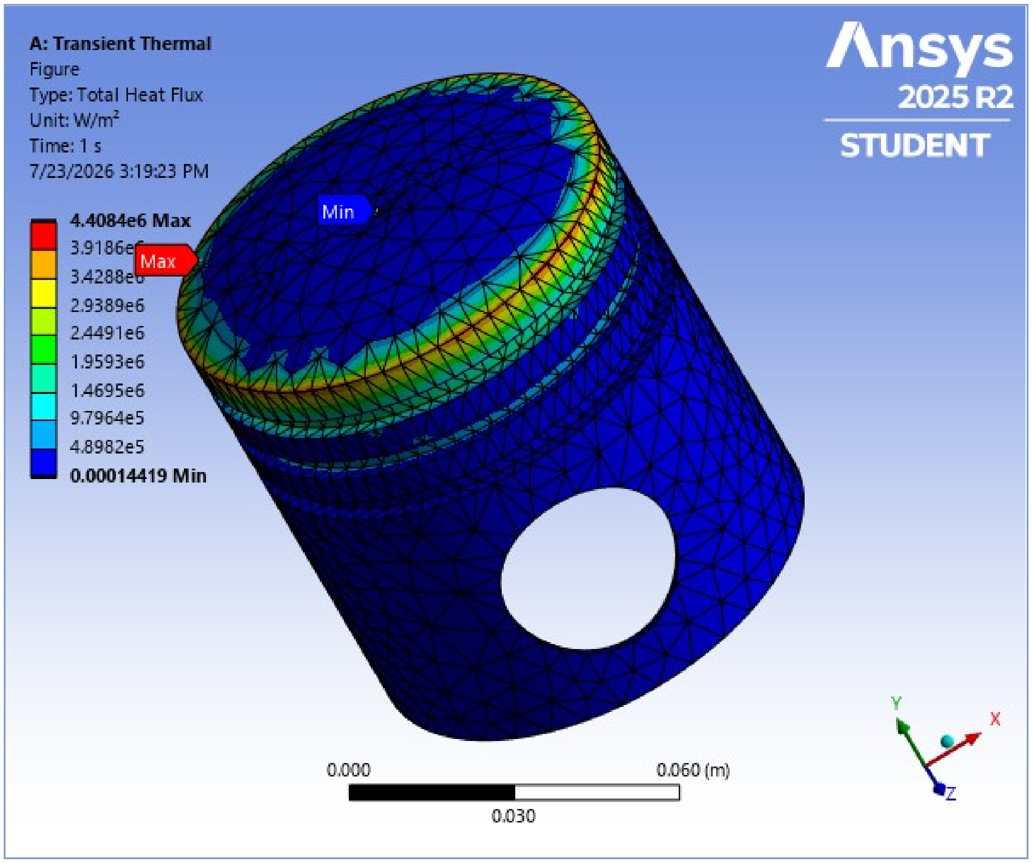


TABLE 19
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux

Time [s]	Minimum [W/m²]	Maximum [W/m²]	Average [W/m²]
1.e-002	1.4419e-004	4.4084e+006	6.1577e+005
2.e-002			
5.e-002			
0.14			
0.24			
0.34			
0.44			
0.54			
0.64			
0.74			
0.84			
0.94			
1.			

FIGURE 9
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux > Figure



Static Structural (B5)

TABLE 20
Model (A4, B4) > Analysis

Object Name	*Static Structural (B5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 21
Model (A4, B4) > Static Structural (B5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	3.
Current Step Number	2.
Step End Time	2. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	

Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\User\OneDrive\Desktop\Piston\Ansys\Ansys\d\d_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 22
Model (A4, B4) > Static Structural (B5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time
1	1. s
2	2. s
3	6. s

TABLE 23
Model (A4, B4) > Static Structural (B5) > Loads

Object Name	*Fixed Support*	*Pressure*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	2 Faces	3 Faces
Definition		
Type	Fixed Support	Pressure
Suppressed	No	
Define By		Normal To
Applied By		Surface Effect
Loaded Area		Deformed

Magnitude		Tabular Data
Tabular Data		
Independent Variable		Time

FIGURE 10
Model (A4, B4) > Static Structural (B5) > Pressure

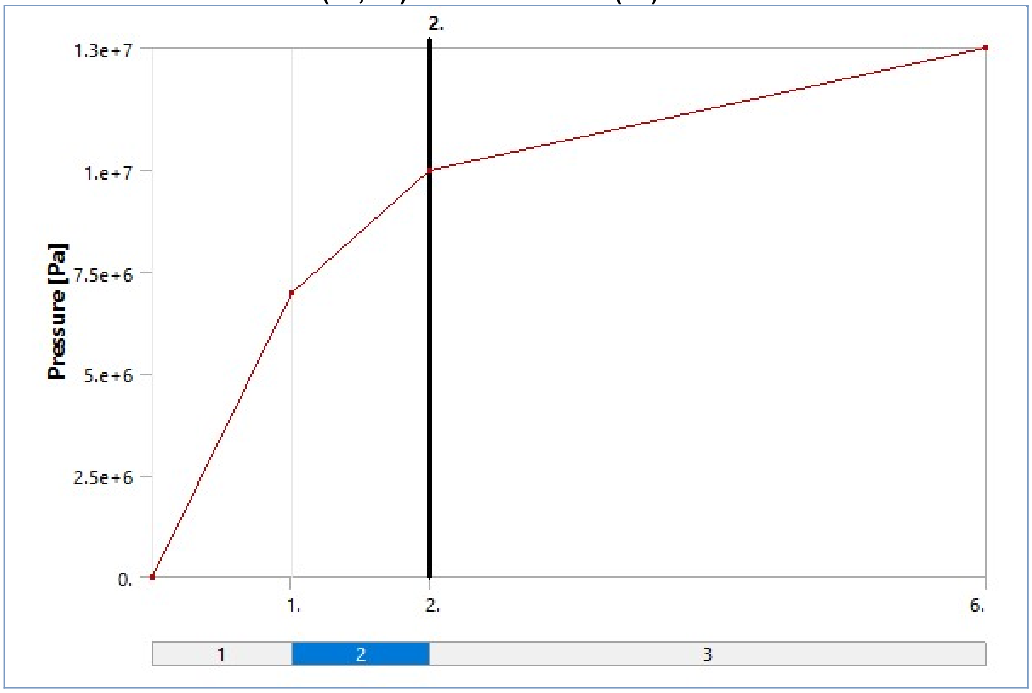


TABLE 24
Model (A4, B4) > Static Structural (B5) > Pressure

Steps	Time [s]	Pressure [Pa]
1	0.	0.
	1.	7.e+006
2	2.	1.e+007
3	6.	1.3e+007

TABLE 25
Model (A4, B4) > Static Structural (B5) > Imported Load (A6)

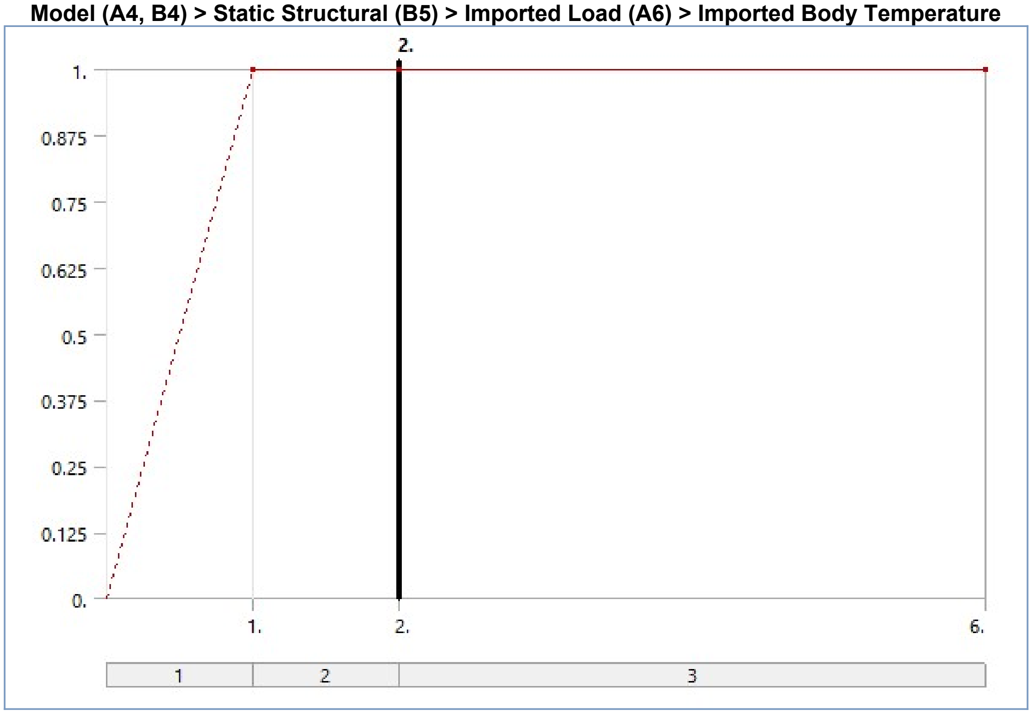
Object Name	*Imported Load (A6)*
State	Fully Defined
Definition	
Type	Imported Data
Interpolation Type	Mechanical Results Transfer
Suppressed	No
Source	A6::Solution 2
Data Management	
Delete Mapped Data Files	Yes

TABLE 26
Model (A4, B4) > Static Structural (B5) > Imported Load (A6) > Imported Body Temperature

Object Name	*Imported Body Temperature*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Type	Imported Body Temperature
Tabular Loading	Program Controlled
Mapped Data	To Input File
Suppressed	No

Source Environment	Transient Thermal (A5)
Source Time	Worksheet

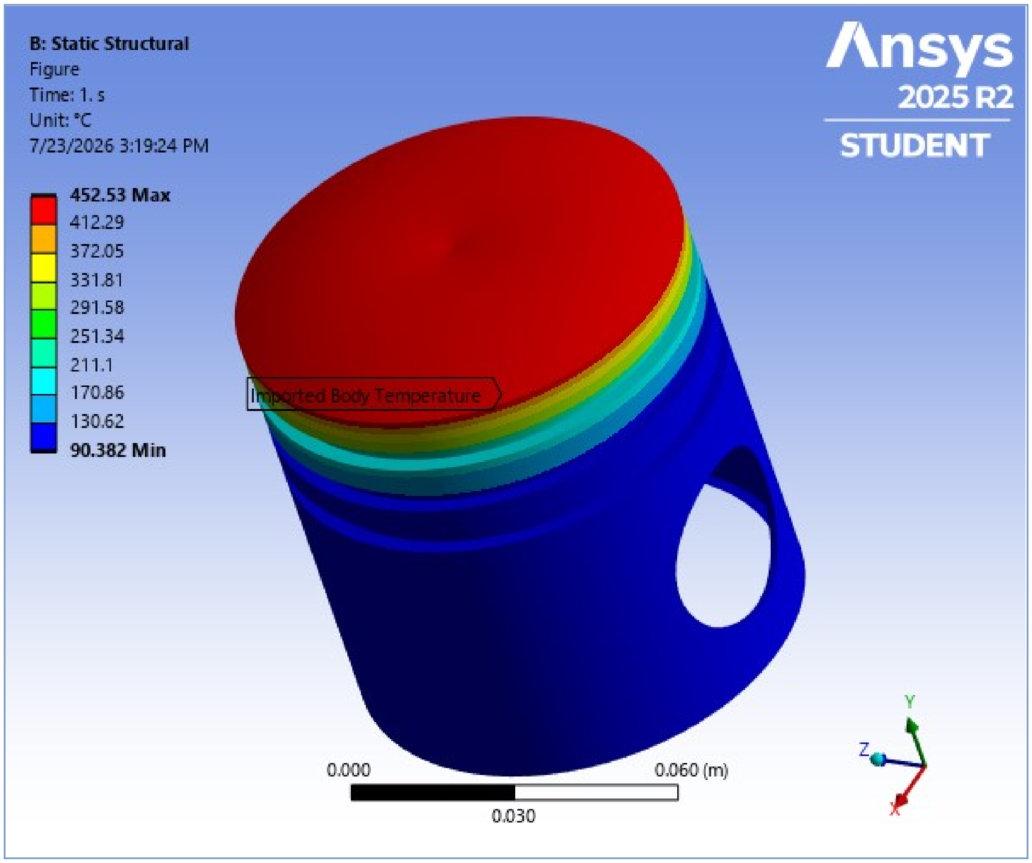
FIGURE 11
Model (A4, B4) > Static Structural (B5) > Imported Load (A6) > Imported Body Temperature



Model (A4, B4) > Static Structural (B5) > Imported Load (A6) > Imported Body Temperature

	Source Time (s)	Analysis Time (s)	Scale	Offset (Δ°C)
1	End Time	1	1	0
*				

FIGURE 12
Model (A4, B4) > Static Structural (B5) > Imported Load (A6) > Imported Body Temperature > Figure



Solution (B6)

TABLE 27
Model (A4, B4) > Static Structural (B5) > Solution

Object Name	*Solution (B6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	27. s
MAPDL Memory Used	284. MB
MAPDL Result File Size	13.812 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 28
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes

Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 29
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Results

Object Name	*Total Deformation*	*Equivalent Stress*	*Equivalent Elastic Strain*
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Type	Total Deformation	Equivalent (von-Mises) Stress	Equivalent Elastic Strain
By	Time		
Display Time	Last		
Separate Data by Entity	No		
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Results			
Minimum	0. m	6.7352e+006 Pa	5.7448e-004 m/m
Maximum	7.867e-003 m	2.722e+009 Pa	3.7198e-002 m/m
Average	8.6186e-004 m	5.1341e+008 Pa	9.8595e-003 m/m
Minimum Occurs On	SYS-2\Solid		
Maximum Occurs On	SYS-2\Solid		
Minimum Value Over Time			
Minimum	0. m	3.0172e+006 Pa	3.0213e-004 m/m
Maximum	0. m	6.7352e+006 Pa	5.7448e-004 m/m
Maximum Value Over Time			
Minimum	4.092e-003 m	1.4614e+009 Pa	1.9971e-002 m/m
Maximum	7.867e-003 m	2.722e+009 Pa	3.7198e-002 m/m
Information			
Time	6. s		
Load Step	3		
Substep	1		
Iteration Number	3		
Integration Point Results			
Display Option		Averaged	
Average Across Bodies		No	

FIGURE 13
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Total Deformation

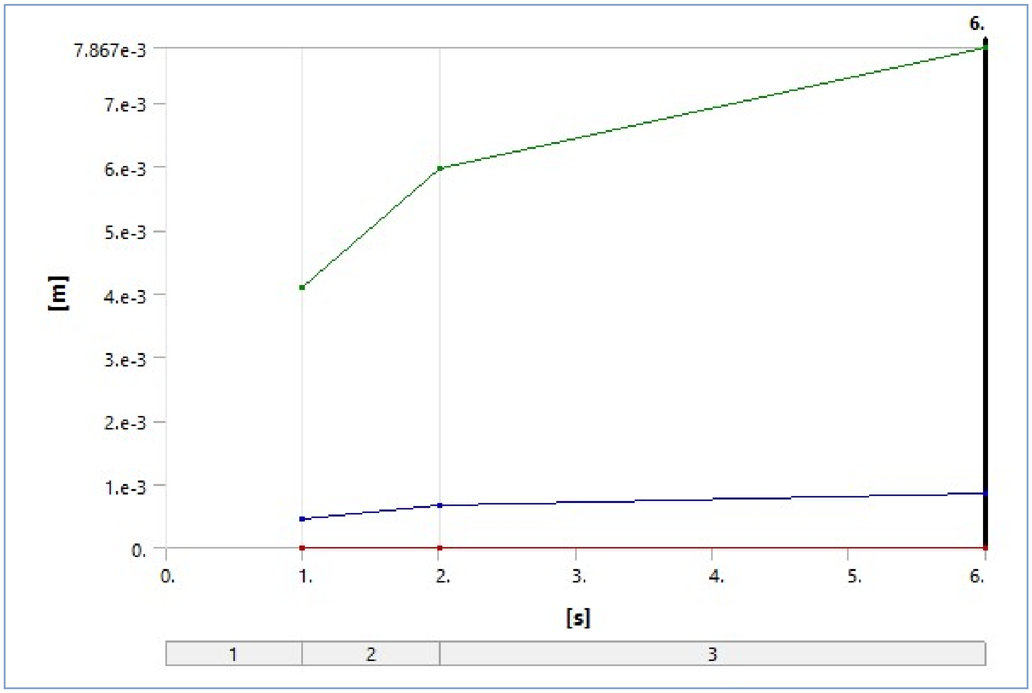


TABLE 30
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	0.	4.092e-003	4.614e-004
2.		5.9787e-003	6.5966e-004
6.		7.867e-003	8.6186e-004

FIGURE 14
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Total Deformation > Figure

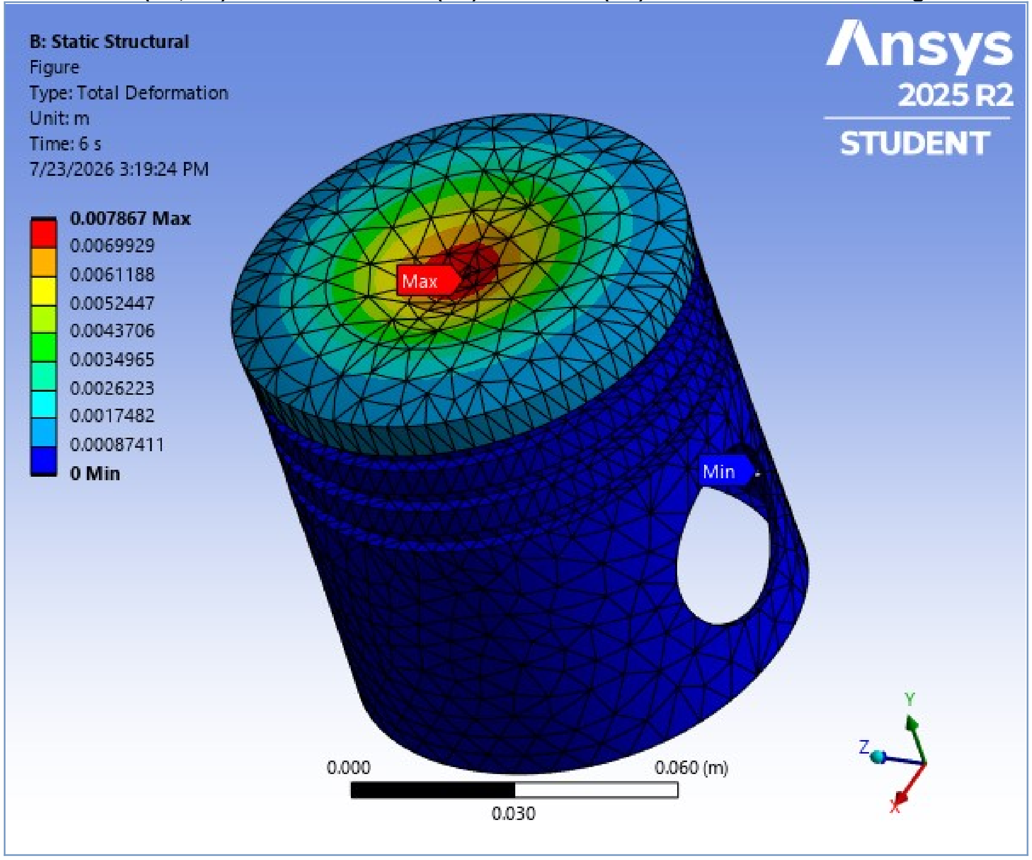


FIGURE 15
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

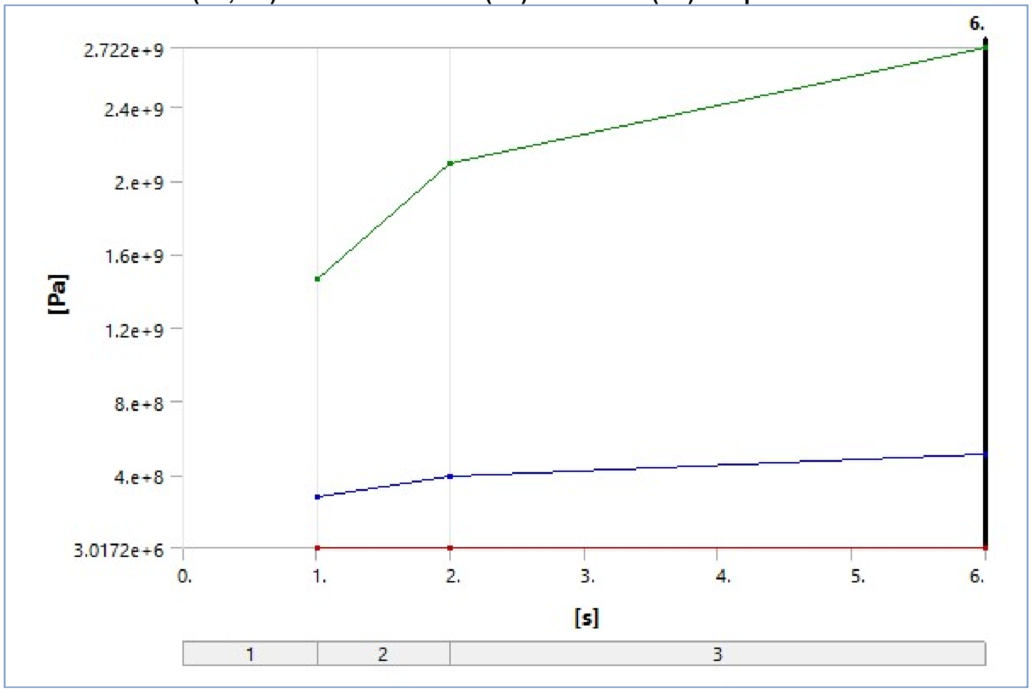


TABLE 31
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	3.0172e+006	1.4614e+009	2.7858e+008
2.	4.8668e+006	2.0917e+009	3.9587e+008
6.	6.7352e+006	2.722e+009	5.1341e+008

FIGURE 16
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress > Figure

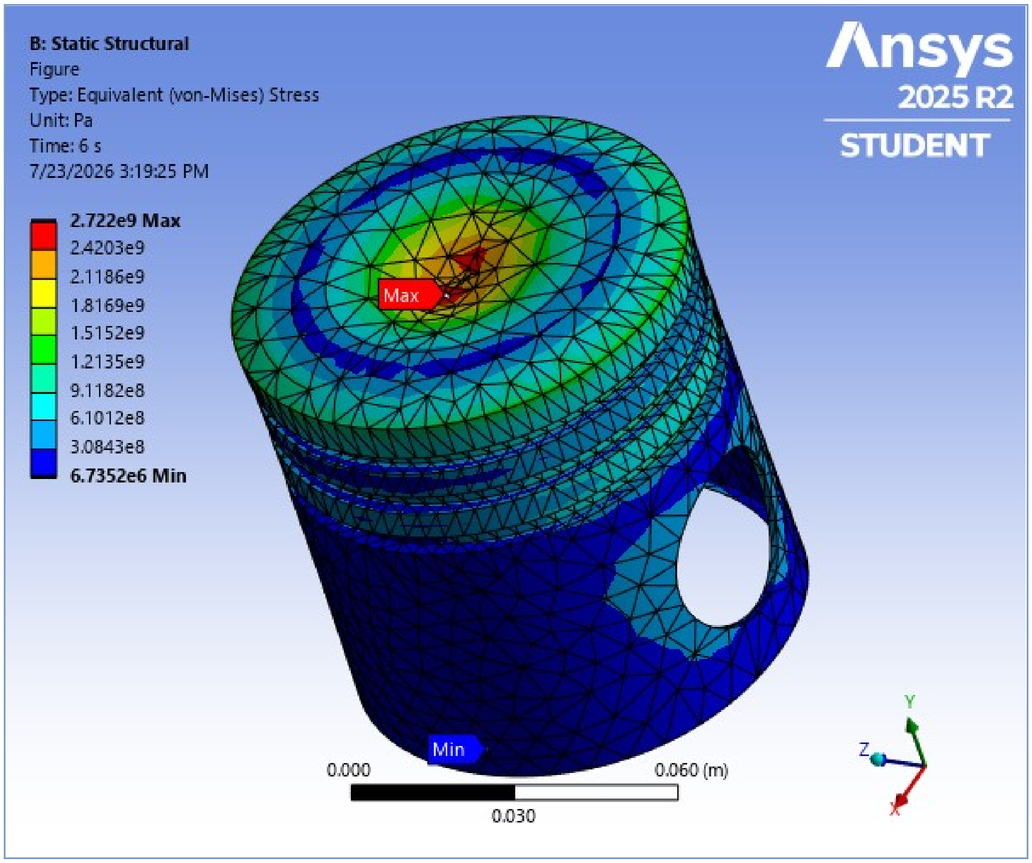


FIGURE 17
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Equivalent Elastic Strain

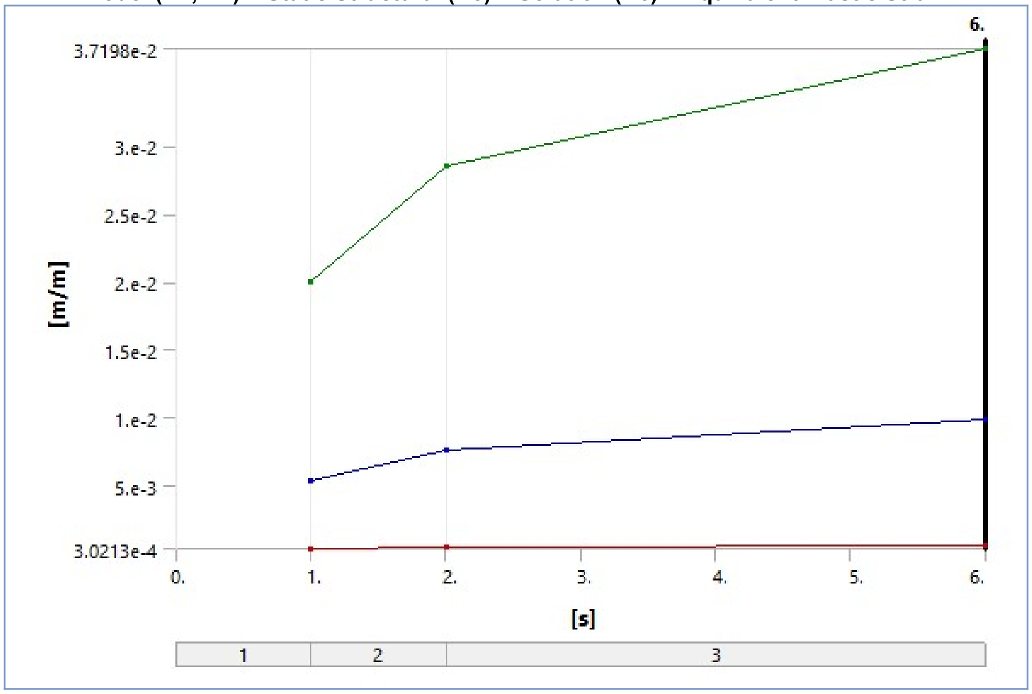
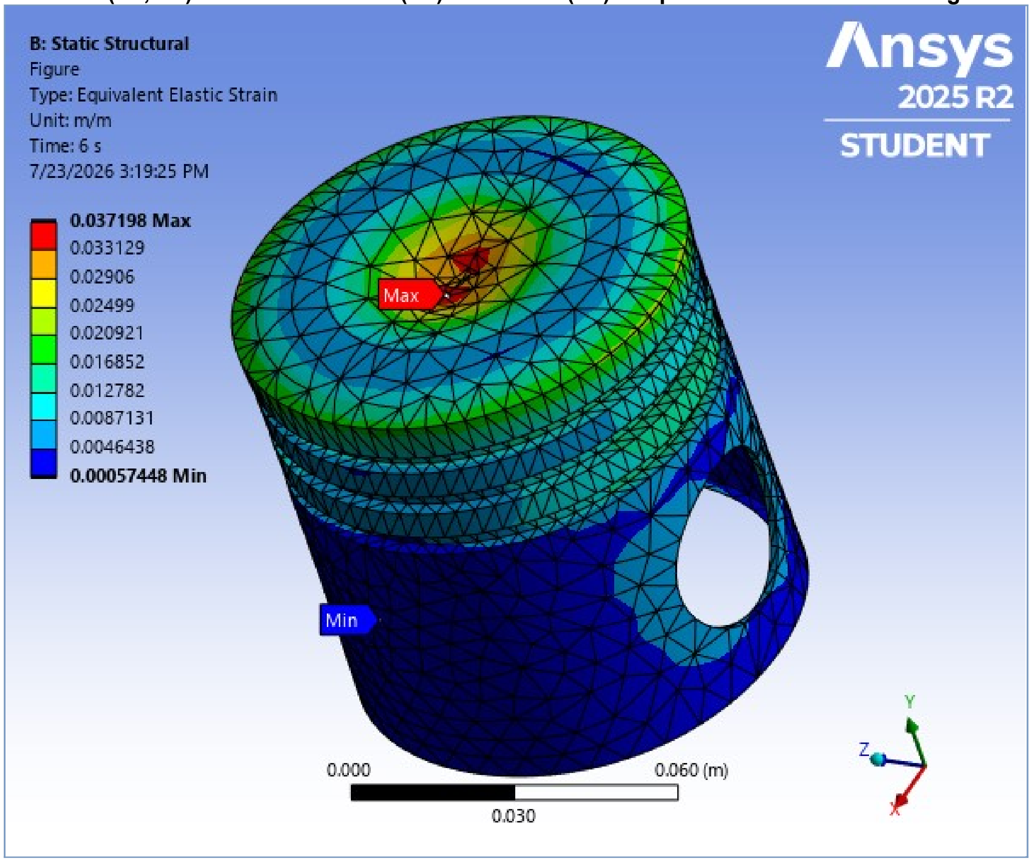


TABLE 32
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Equivalent Elastic Strain

Time [s]	Minimum [m/m]	Maximum [m/m]	Average [m/m]
1.	3.0213e-004	1.9971e-002	5.3572e-003
2.	4.3718e-004	2.8585e-002	7.6063e-003
6.	5.7448e-004	3.7198e-002	9.8595e-003

FIGURE 18
Model (A4, B4) > Static Structural (B5) > Solution (B6) > Equivalent Elastic Strain > Figure



Material Data

Al2618

TABLE 33
Al2618 > Constants

Density	2770 kg m^-3
Thermal Conductivity	147 W m^-1 C^-1
Coefficient of Thermal Expansion	2.22e-005 C^-1

TABLE 34
Al2618 > Color

Red	Green	Blue
170	170	170

TABLE 35
Al2618 > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
7.35e+010	0.33	7.2059e+010	2.7632e+010	

TABLE 36
Al2618 > Tensile Yield Strength

Tensile Yield Strength Pa
3.72e+008

TABLE 37
Al2618 > Compressive Yield Strength

Compressive Yield Strength Pa
3.72e+008

TABLE 38
Al2618 > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
4.41e+008

TABLE 39
Al2618 > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
4.41e+008